

College Council & Club Management System

CSE-302: Database Management System

Relational Schema Report

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Date: August 24, 2026

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1 Overview

This report presents the relational schema derived from the Entity-Relationship diagram of the College Council & Club Management System. Each schema definition marks its primary key and foreign key attributes, and states the Relationship Integrity Constraint (RIC) enforced by each foreign key.

2 ER-to-Relational Mapping Rules Applied

- **Strong entity** $\rightarrow$ a table, with the entity's key attribute(s) becoming the primary key.
- **1:N relationship (no relationship attributes)** $\rightarrow$ no separate table; the primary key of the “1” side is placed as a foreign key on the “N” side (e.g., *Contains* becomes `CouncilID` as a foreign key in `Club`).
- **M:N relationship carrying its own attributes** $\rightarrow$ a separate table, whose primary key is a new surrogate key (or the composite of the participating entities' keys), containing foreign keys to every participating entity plus the relationship's own attributes (e.g., *Membership* becomes the `ClubMembership` table).

3 Relational Schema

Underlined attributes denote primary keys; attributes in *italics* denote foreign keys. Each foreign key's Relationship Integrity Constraint (RIC) is shown as an arrow from the foreign key attribute to the primary key it references.

3.1 Council

Council (`CouncilID`, Name, Description)

- No foreign keys; no RIC.

3.2 Club

Club (`ClubID`, Name, *`CouncilID`*)

- **RIC:** `Club.CouncilID` $\longrightarrow$ `Council.CouncilID`

3.3 Vertical

Vertical (`VerticalID`, Name, *`ClubID`*)

- **RIC:** `Vertical.ClubID` $\longrightarrow$ `Club.ClubID`

3.4 Event

Event (`EventID`, Name, EventDate, AcademicYear, *`ClubID`*)

- **RIC:** `Event.ClubID` $\longrightarrow$ `Club.ClubID`

3.5 Student

Student (RollNumber, Name, GraduationYear, Branch, Email)

- No foreign keys; no RIC.

3.6 Room

Room (RoomID)

- No foreign keys; no RIC.

3.7 ClubMembership

ClubMembership (MembershipID, *RollNumber*, *ClubID*, PORTitle, AcademicYear)

- **RIC 1:** ClubMembership.RollNumber $\rightarrow$ Student.RollNumber
- **RIC 2:** ClubMembership.ClubID $\rightarrow$ Club.ClubID

3.8 ClubRoomAllocation

ClubRoomAllocation (AllocationID, *ClubID*, *RoomID*, AcademicYear)

- **RIC 1:** ClubRoomAllocation.ClubID $\rightarrow$ Club.ClubID
- **RIC 2:** ClubRoomAllocation.RoomID $\rightarrow$ Room.RoomID

3.9 EventInvolvement

EventInvolvement (InvolvementID, *RollNumber*, *EventID*, RoleCategory, SpecificRole, ResultDetails)

- **RIC 1:** EventInvolvement.RollNumber $\rightarrow$ Student.RollNumber
- **RIC 2:** EventInvolvement.EventID $\rightarrow$ Event.EventID

4 Conclusion

The relational schema above maps each strong entity from the ER diagram to its own table, resolves the three one-to-many relationships (*Contains*, *Has*, *Organizes*) by placing foreign keys on the “many” side, and resolves the three many-to-many relationships carrying their own attributes (*Membership*, *Allocated*, *Involved*) into separate mapping tables. Every foreign key is accompanied by its Relationship Integrity Constraint, ensuring that no orphaned records (e.g., a club without a valid council, or an event without a valid club) can exist in the database.